

南アジア古典学

2026

第21号

目次

<i>Kalpalatā</i> と <i>Avadānamālā</i> の研究 (14) — TJAM 第 20 章～第 26 章 —	岡野 潔	1
ある律典の注釈書に属する未同定の 3 葉	生野 昌範	293
ネパールの演劇写本 ジャガトプラカーシャ・マッラ王のネワール語歌集 (9)	北田 信	305
インド伝統医学における日常的な養生法 —ネパール版 <i>Suśrutasamhitā</i> 第 4 篇第 24 章校訂・和訳研究—	榮 惟樹	313
初期アドヴァイタ学派における顕現説と映像説の萌芽	梶野 歩夢	361
マンダナミシュラの行為論： <i>Bhāvanāviveka</i> 前論部 (Pūrvapakṣa) 訳注研究	斉藤 茜	381
Stemmatic Analysis of the Manuscripts of <i>Tantravārttika</i> ad 2.1.46	Kei KATAOKA	487
How can an exclusion be known and established? —Kumārila's critique of <i>apoha</i> and Śāntarakṣita's reply	LongYin SIN	495

九州大学大学院人文科学府・文学部

インド哲学史研究室

Stemmatic Analysis of the Manuscripts of *Tantravārttika* ad 2.1.46

Kei Kataoka

1. Introduction

This paper is a companion to a forthcoming critical edition of Kumārila's *Tantravārttika* ad 2.1.46 (*Ekavākyatālakṣaṇādhikaraṇa*) (Kataoka forthcoming). Its primary aim is to present, by means of a stemma codicum, the relationships among the twelve manuscripts and an earlier printed edition employed in that edition.

The analysis was conducted with the assistance of Claude, a generative AI developed by Anthropic — in particular, the recently released Opus 4.6 model. A separate paper was deemed appropriate for this stemmatic analysis, given the level of detail that Claude's output involves, which would have rendered it unwieldy as part of the edition's introduction proper.

In my earlier critical editions of *Tantravārttika* 2.1.1–4 (Kataoka 2004) and 3.1.1 (Kataoka 2024), stemmas were included, but they were largely impressionistic — rough conceptual diagrams grounded in accumulated experience rather than explicit quantitative reasoning. The present paper, by contrast, offers a statistically grounded analysis based on concrete numerical data.

The immediate occasion for this paper was the following: when I fed the working files for the forthcoming edition — specifically the .dn file (a pre-TeX intermediate format) and the associated .sty file — into Claude, together with supplementary manuscript information, the results of the stemmatic analysis were, unexpectedly, remarkably clean. Sharing that output is thus a secondary purpose of this paper.

More broadly, a further aim is to demonstrate that generative AI can be a genuinely effective tool for stemmatic analysis of Sanskrit manuscript traditions. In this sense, the paper is intended as a practical case study in the application of generative AI to Sanskrit philology. Claude itself is a rapidly developing system, and the use of generative AI for manuscript stemmatic analysis is, to the author's knowledge, without precedent. There are no doubt methodological issues to be identified and discussed. For the present, however, the goal is simply to share the results and to take a first step toward what may prove a productive new direction in the field.

2. Consulted Witnesses

The following table summarises the twelve manuscript witnesses and the printed edition consulted for my forthcoming critical edition, with dates where known.

Siglum	Repository	Script	Date	Notes
A ⁴	Ānandāśrama ed.	—	1929–1934	Based on editio princeps (A, 1882–1903)

B ₁	BORI, Pune	Devanāgarī	1493 CE (Śaka 1415)	No. 93 of 1869–70. 138 folios. Oldest dated ms.
B ₂	BORI, Pune	Devanāgarī	1868 CE (Śaka 1789)	No. 101 of 1860–68. 437 folios. Late copy.
G ₁	Asiatic Society, Kolkata	Devanāgarī	1638 CE (?)	G. 1256. 48 folios. 2 folios.
H ₁	Osmania Univ., Hyderabad	Devanāgarī	17th c. CE	B. 102/2. 68 folios.
I ₁	British Library, London	Devanāgarī	undated	IO 1451a (No. 2153). 62 folios.
I ₄	British Library, London	Grantha	1860 CE	Burnell 531 (No. 5899). 578 pp.
O ₁	Bodleian Library, Oxford	Devanāgarī	1542 CE (saṃvat 1598)	Wilson 273b (No. 522). 58 folios.
O ₂	Bodleian Library, Oxford	Devanāgarī	early 16th c.	Wilson 325 (No. 520). 25 folios.
R ₁	Royal Asiatic Society, London	Malayāḷam	1831 CE	Whish MS 107. Palm leaf.
T ₁	Sarasvati Mahal, Thanjavur	Devanāgarī	undated	No. 6770. 276 folios. Undated; apparently old.
T ₂	Sarasvati Mahal, Thanjavur	Devanāgarī	c. 1650 CE	No. 6774. Kavīndrācārya lineage. T ₂ →I ₂ apograph.
W ₁	Prajñā Pāṭhaśālā, Wai	Devanāgarī	1563 CE (Śaka 1485)	No. 6191. 82 folios.

3. Methodology

All 253 variant-reading entries in the apparatus were parsed computationally. For each entry, every witness was assigned to a reading-group: group 0 (= lemma) or groups 1, 2, etc. (= variant readings). The following metrics were then calculated:

Pairwise distance: percentage of entries in which two witnesses carry different readings (when both are present).

Shared non-lemma readings: count of entries where two or more witnesses share the same non-lemma reading.

Exclusive shared errors: cases where a group of witnesses shares a non-lemma reading that no other witness shares. These are the decisive criterion for stemmatic grouping.

4. Pairwise Distance Matrix (%)

The table below shows the percentage of disagreement between each pair of witnesses. Lower values indicate closer textual affinity. Cells are shaded from green (close) to red (distant).

	A ⁴	B ₁	B ₂	G ₁	H ₁	I ₁	I ₄	O ₁	O ₂	R ₁	T ₁	T ₂	W ₁
A ⁴	-	17.0	14.6	12.4	17.8	5.5	46.1	11.5	13.4	32.4	57.9	17.4	25.9
B ₁	17.0	-	20.9	14.0	20.2	13.4	42.0	15.0	13.0	30.4	53.2	24.9	25.5
B ₂	14.6	20.9	-	18.3	22.1	11.9	49.4	16.6	18.2	35.6	59.1	24.1	29.9
G ₁	12.4	14.0	18.3	-	15.1	9.7	40.4	13.4	13.4	25.3	52.4	18.3	25.0
H ₁	17.8	20.2	22.1	15.1	-	15.0	43.3	17.4	16.2	29.6	56.7	22.5	27.9
I ₁	5.5	13.4	11.9	9.7	15.0	-	42.4	7.5	9.9	28.1	53.6	15.8	23.1
I ₄	46.1	42.0	49.4	40.4	43.3	42.4	-	44.1	43.3	38.4	37.3	44.1	52.3
O ₁	11.5	15.0	16.6	13.4	17.4	7.5	44.1	-	11.5	28.5	54.8	19.4	24.7
O ₂	13.4	13.0	18.2	13.4	16.2	9.9	43.3	11.5	-	26.9	54.8	20.9	19.9
R ₁	32.4	30.4	35.6	25.3	29.6	28.1	38.4	28.5	26.9	-	47.2	31.2	39.0
T ₁	57.9	53.2	59.1	52.4	56.7	53.6	37.3	54.8	54.8	47.2	-	55.2	61.6
T ₂	17.4	24.9	24.1	18.3	22.5	15.8	44.1	19.4	20.9	31.2	55.2	-	33.9
W ₁	25.9	25.5	29.9	25.0	27.9	23.1	52.3	24.7	19.9	39.0	61.6	33.9	-

Key observations: T₁ and I₄ are the most deviant witnesses overall (distance >40% from almost all others), yet they are closest to each other (37.3%). R₁ is intermediate. I₁ and A⁴ are the closest pair (5.5%), followed by I₁–O₁ (7.5%). This pattern strongly suggests a bipartite stemma.

5. Exclusive Shared Errors

Exclusive shared errors (innovations shared by a group of witnesses and no others) are the primary evidence for establishing stemmatic relationships. The following table lists all attested exclusive pairings and groupings.

Witness Group	Exclusive	Shared (total)	Interpretation
T ₁ –I ₄	25	43	Strongest pair. Confirms shared ancestor δ (siblings, not parent-child).
T ₁ –I ₄ –R ₁	7	n/a	Shared ancestor of all three (α -hyparchetype).
R ₁ –T ₁ (excl. I ₄)	6	17	Sub-grouping within α : R ₁ closer to T ₁ than to I ₄ .
T ₁ –W ₁	3	6	Possible contamination from α into W ₁ .
B ₁ –G ₁	2	15	Shared readings within β ; minor sub-group.
B ₁ –O ₁ –O ₂ –W ₁	2	n/a	Western β sub-group.
H ₁ –R ₁	2	8	Possible contamination between families.
O ₂ –W ₁	1	28	Close textual affinity (19.9% distance).
H ₁ –W ₁	1	21	Minor link within β .
I ₄ –R ₁	1	13	Supports α grouping.
G ₁ –R ₁	1	3	Possible contamination.

6. Family Structure: α and β

6.1 The α -family (Southern recension)

The witnesses T_1 , I_4 , and R_1 constitute a distinct family (α). The evidence is overwhelming:

(a) T_1 and I_4 share 25 exclusive errors, confirming a close genetic relationship. However, T_1 carries 50 unique innovations absent from I_4 , and I_4 carries 24 unique innovations absent from T_1 , ruling out a direct parent-child relationship. They are siblings descending from a common ancestor δ . T_1 is probably the oldest surviving manuscript (15th c.?), while I_4 (1860 CE, Grantha script) is a late but relatively faithful copy of a δ -type exemplar.

(b) When the α -group deviates from the lemma, 90 entries show α deviating alone vs. 89 where only β deviates, confirming a clean bipartite split.

(c) R_1 , a Malayālam palm-leaf manuscript from 1831, shares 7 exclusive errors with the T_1 – I_4 pair and 6 additional exclusive errors with T_1 alone, placing it firmly in the α -family while suggesting that R_1 and T_1 may share a more recent common ancestor within α .

(d) I_4 agrees with T_1 in 39.1% of T_1 's deviations from the lemma. R_1 agrees 14.7%. No β -witness exceeds 7.8%.

6.2 The β -family (Northern recension)

All remaining witnesses (A^4 , B_1 , B_2 , G_1 , H_1 , I_1 , O_1 , O_2 , T_2 , W_1) belong to the β -family. Within β , the sub-structure is less sharply defined, suggesting a more bushy transmission:

(a) I_1 and A^4 are nearly identical (5.5% distance in TV 2.1.46; 8.7% in TV 2.1.1–4), indicating that A (the editio princeps, on which A^4 is based) was based on a manuscript very close to I_1 or its exemplar.

(b) O_1 is very close to I_1 (7.5%), and O_2 to O_1 (11.5%), forming a compact sub-group: I_1 – O_1 – O_2 .

(c) G_1 occupies a relatively independent position within β , with moderately low distances to most β -witnesses (9.7–18.3%). It shares 2 exclusive errors with B_1 .

(d) T_2 is unambiguously a β -witness (T_2 shares 0 exclusive deviations with α but 34 non-lemma readings with β alone), despite being housed in Thanjavur alongside T_1 . T_2 derives from the Kavīndrācārya manuscript through a β -lineage, and I_2 is its apograph ($T_2 \rightarrow I_2$).

(e) W_1 shows the highest distance from β -core witnesses (23–34% distance) and 3 exclusive shared errors with T_1 (α). This suggests possible contamination from an α -source.

6.3 Evidence for contamination

Several indicators suggest that the manuscript tradition is not purely bifurcating:

(a) W_1 shares 3 exclusive readings with T_1 (α) despite being a β -witness. W_1 (dated 1563 CE, Wai, Maharashtra) may have been corrected against a Southern exemplar.

(b) H₁ shares 2 exclusive readings with R₁ (α). This cross-family agreement, although limited, may indicate sporadic contamination.

(c) R₁ (Malayālam script, 1831) shows intermediate distance values (25–39% to β , 38–47% to T₁/I₄), consistent with its position as an α -witness having been exposed to β -readings.

7. Chronological Alignment

The following timeline integrates manuscript dates with the proposed stemma:

Date	Witness	Family	Significance
1493 CE	B ₁ (BORI 93)	β	Oldest dated ms. Devanāgarī, Pune.
1542 CE	O ₁ (Bodleian 273b)	β	Northern Indian provenance.
early 16th c.	O ₂ (Bodleian 325)	β	Close to O ₁ (11.5% distance).
1563 CE	W ₁ (Wai 6191)	β (contam.)	Shows contamination from α . Maharashtra.
1638 CE (?)	G ₁ (Asiatic Soc.)	β	Independent β -witness.
c. 1650 CE	T ₂ (Thanjavur 6774)	β	Kavīndrācārya lineage. T ₂ →I ₂ apograph.
17th c.	H ₁ (Osmania)	β	Hyderabad. Minor α -contamination.
undated	T ₁ (Thanjavur 6770)	α	Southern recension.
undated	I ₁ (BL 2153)	β	Closest to A ⁴ (5.5% distance).
1831 CE	R ₁ (RAS, Whish 107)	α	Malayālam palm leaf. Late α -copy.
1860 CE	I ₄ (BL, Burnell 531)	α	Grantha script. Late α -copy of a δ -type exemplar.
1868 CE	B ₂ (BORI 101)	β	Latest dated ms. Kolhapur.
1929–34	A ⁴ (Ānandāśrama)	β	Based on A, which derives from an I ₁ -type ms.

Legend: Green = β -family; Pink = α -family; Orange = contaminated.

8. Proposed Stemma Codicum

Based on the analysis above, the following stemma is proposed. Solid lines represent genetic descent; dashed lines represent contamination.

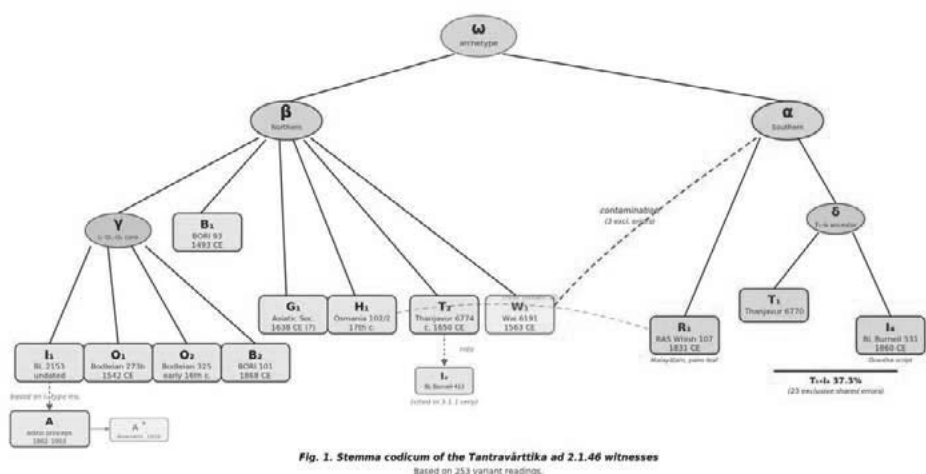


Figure 1. Proposed stemma codicum for TV 2.1.46. ω = archetype; α = Southern hyparchetype (T₁, I₄, R₁); δ = common ancestor of T₁ and I₄ (25 exclusive shared errors); β = Northern hyparchetype (B₁, B₂, G₁, H₁, I₁, O₁, O₂, T₂, W₁); γ = sub-group comprising I₁, O₁, O₂, and B₂. A (editio princeps, 1882–1903) is based on an I₁-type manuscript. A* is another edition based on A. Dashed lines from W₁ and H₁ indicate contamination from the α -family. T₂ → I₂ is an apograph relationship (I₂ cited only in TV 3.1.1).

9. Summary of Findings

Finding	Evidence
Bipartite stemma: α vs. β	α deviates alone 90×; β deviates alone 89×; both deviate differently 68×.
T ₁ –I ₄ closest within α	25 exclusive shared errors; 37.3% distance. Siblings via common ancestor δ (T ₁ : 50 unique errors; I ₄ : 24).
T ₂ →I ₂ apograph confirmed	T ₂ is β (0 exclusive α -agreements); I ₄ is α . Different transmission lines despite same library.
R ₁ is α with own innovations	7 exclusive T ₁ –I ₄ –R ₁ errors; R ₁ intermediate distance (38–47% to core α).
A ≈ I ₁	5.5% distance (TV 2.1.46); 8.7% (TV 2.1.1–4). Editio princeps based on I ₁ -type exemplar.
β_2 sub-group: I ₁ –O ₁ –O ₂ –B ₂	Pairwise distances 7.5–18.2%. O ₁ –O ₂ = 11.5%.
W ₁ contaminated	3 exclusive T ₁ –W ₁ readings; highest β -distance (23–34%).
H ₁ minor contamination	2 exclusive H ₁ –R ₁ readings; otherwise firmly β .
No β sub-group strongly attested	No exclusive group of ≥3 β -witnesses found. Bushy transmission within β .

10. Editorial Implications

The bipartite structure (α/β) means that agreement of α and β against any single witness is strong evidence for the archetype reading. In practice:

(1) When α ($T_1/I_4/R_1$) and the majority of β agree, the reading is almost certainly archetypal.

(2) When α and β disagree, editorial judgment is required. Neither family is inherently superior; α preserves some genuine readings lost in β , and vice versa.

(3) Within β , I_1 and A should not be counted as two independent witnesses, since A is based on an I_1 -type exemplar.

(4) W_1 's agreement with α against other β -witnesses should be evaluated for possible contamination rather than taken as independent confirmation.

Bibliography

Kataoka, Kei

2004 *Koten Indo no Saishiki Kōi-ron: Sābarabhāṣya & Tantravārttika ad 2.1.1–4 Genten Kōtei Yakuchū Kenkyū*. [The Theory of Ritual Action in Mīmāṃsā: Critical Edition and Annotated Japanese Translation of Sābarabhāṣya & Tantravārttika ad 2.1.1–4.] Tokyo: Sankibō Press.

2024 “A Critical Edition of Kumārila’s *Tantravārttika* ad 3.1.1: *Śeṣapratijñādhikaraṇa*.” *Tōyō Bunka Kenkyūjo Kiyō* (The Memoirs of the Institute for Advanced Studies on Asia) 186, 130(111)–80(155).

forthcoming “A Critical Edition of Kumārila’s *Tantravārttika* ad 2.1.46: *Ekavākyatālakṣaṇādhikaraṇa*.” *Tōyō Bunka Kenkyūjo Kiyō* (The Memoirs of the Institute for Advanced Studies on Asia).

SOUTH ASIAN CLASSICAL STUDIES

No. 21

2026

CONTENTS

A Study of the <i>Avadānakalpalatā</i> and the <i>Avadānamālās</i> (14) —TJAM Chapters 20–26—	OKANO Kiyoshi	1
Three Unidentified Folios of a Vinaya Commentary	SHŌNO Masanori	293
Dramatic Manuscripts from Nepal: Jagatprakāśa Malla's Newari Songs (9)	KITADA Makoto	305
Daily Health Regimens in Indian Traditional Medicine: A Critical Edition and Japanese Translation of the Nepalese Version of <i>Suśrutasamhitā</i> 4.24	SAKAE Yuiki	313
The Development of <i>Ābhāsavāda</i> and <i>Pratibimbavāda</i> in the Early Advaita Vedānta	KAJINO Ayumu	361
Maṇḍana Miśra's Theory of Action: An Annotated Japanese Translation of the <i>Bhāvanāviveka</i> Pūrvapakṣa	SAITO Akane	381
Stemmatic Analysis of the Manuscripts of <i>Tantravārttika</i> ad 2.1.46	KATAOKA Kei	487
How can an exclusion be known and established? —Kumārila's critique of <i>apoha</i> and Śāntarakṣita's reply	SIN LongYin	495